DATA STRUCTURE
LAB PROGRAM

24. Implementation of Minimum Spanning Tree using Prim's Algorithm

```
#include <stdio.h>

#define V 5
#define INF 9999

int main()
{
    int graph[V][V] = {
        {0, 2, 0, 6, 0},
        {2, 0, 3, 8, 5},
        {0, 3, 0, 0, 7},
        {6, 8, 0, 0, 9},
        {0, 5, 7, 9, 0}
    };

    int selected[V] = {0};
    int no_edge = 0;
    int x, y;
    int totalCost = 0;
```

```
selected[0] = 1;

printf("Edges in the Minimum Spanning Tree:\n");
printf("Edge\tWeight\n");

while (no_edge < V - 1)
{
    int min = INF;
    x = 0;
    y = 0;

    for (int i = 0; i < V; i++)
    {
        if (selected[i])
        {
            for (int j = 0; j < V; j++)
            {
                if (!selected[j] && graph[i][j])
                {
                    if (min > graph[i][j])
                    {
                        min = graph[i][j];
                        x = i;
                        y = j;
                    }
                }
            }
```

```
            }
        }

        printf("%d - %d\t%d\n", x, y, graph[x][y]);
        totalCost += graph[x][y];

        selected[y] = 1;
        no_edge++;
    }

    printf("\nTotal Cost of MST = %d\n", totalCost);

    return 0;
}
```

Output

```
Edges in the Minimum Spanning Tree:
Edge    Weight
0 - 1   2
1 - 2   3
1 - 4   5
0 - 3   6

Total Cost of MST = 16


=== Code Execution Successful ===
```